

Love that choice 👍

Here are **clean, exam-ready FULL REVISION NOTES** for Chapter 2 (Unit-1: Basics of Python Programming).

You can **read, revise, or even copy this into your notebook.**

CHAPTER 2 – BASICS OF PYTHON PROGRAMMING

FULL REVISION NOTES

◆ 2.1 Declare and Initialize Variables

- **Variable:** Name given to a memory location.
- Python is **dynamically typed** (no need to specify data type).

Syntax:

```
variable_name = value
```

Example:

```
x = 10
```

```
name = "Python"
```

Rules:

- Must start with letter or _
 - No spaces or special characters
 - Case-sensitive
-

◆ 2.2 Comments and Indentation

Comments:

- Used to explain code
- Not executed

Single-line comment

Multi-line (using # or triple quotes):

```
"""
```

This is a comment

```
"""
```

Indentation:

- Defines code blocks
- **Mandatory in Python**
- Standard: **4 spaces**

✗ Wrong indentation → IndentationError

◆ 2.3 Data Types

Basic Built-in Data Types:

Type	Example
------	---------

int	10
-----	----

float	3.14
-------	------

complex	2+3j
---------	------

str	"Hello"
-----	---------

bool	True, False
------	-------------

Check type:

`type(x)`

Type Casting:

`int(), float(), str()`

◆ 2.4 Decision Making Statements

Statements:

- if
- if-else
- if-elif-else
- Nested if

Example:

if marks >= 40:

 print("Pass")

else:

 print("Fail")

Operators:

- Comparison: ==, !=, >, <, >=, <=
 - Logical: and, or, not
-

◆ 2.5 Loop Statements**Types of Loops:**

1. for loop
2. while loop

for loop:

for i in range(1, 6):

 print(i)

while loop:

i = 1

while i <= 5:

 print(i)

 i += 1

Loop Control:

- break
 - continue
 - pass
-

◆ 2.6 Lists, Tuples, Sets, Dictionaries**List []**

- Ordered, mutable, duplicates allowed

```
l = [1, 2, 3]
```

Tuple ()

- Ordered, immutable

```
t = (1, 2, 3)
```

Set {}

- Unordered, no duplicates

```
s = {1, 2, 3}
```

Dictionary {key:value}

- Key–value pairs

```
d = {"name": "Amit", "age": 20}
```

Comparison Table:

Type Ordered Mutable Duplicates

List	Yes	Yes	Yes
Tuple	Yes	No	Yes
Set	No	Yes	No
Dict	Yes	Yes	Keys: No

◆ 2.7 Membership Operators

Operators:

- in
- not in

Example:

```
a = [1, 2, 3]
```

```
print(2 in a)
```

✎ In dictionary, membership checks **keys only**.

◆ 2.8 Strings and Built-in String Functions

String:

- Sequence of characters
- Immutable

```
s = "Python"
```

Indexing:

```
s[0] # P
```

Important Functions:

- upper()
 - lower()
 - capitalize()
 - title()
 - strip()
 - replace()
 - find()
 - count()
 - len()
-

◆ 2.9 Functions (With / Without Arguments)

Function:

Block of reusable code.

```
def function_name():  
    statements
```

Without Arguments:

```
def greet():  
    print("Hello")
```

With Arguments:

```
def add(a, b):
```

return a + b

📌 return sends value back.

💎 2.10 Recursive Method of Problem Solving

- Function calling itself
- Must have **base condition**

Example (Factorial):

```
def fact(n):
```

```
    if n == 0:
```

```
        return 1
```

```
    return n * fact(n-1)
```

✗ No base condition → infinite recursion error

💎 2.11 Recursive vs Iterative Method

Recursive

Iterative

Function calls itself Uses loops

Needs base condition Needs loop condition

More memory usage Less memory usage

Slower Faster

Stack overflow risk No stack issue

✅ FINAL EXAM TIPS

- ✓ Learn definitions
 - ✓ Remember tables (very important)
 - ✓ Practice small programs
 - ✓ Focus on syntax + indentation
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